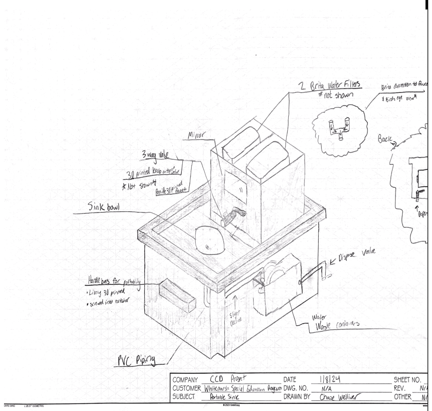
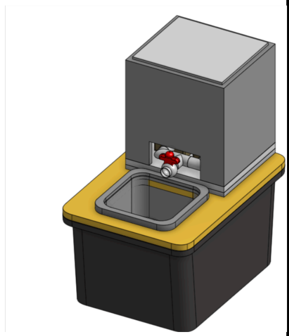
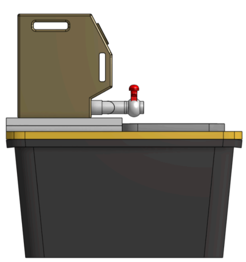
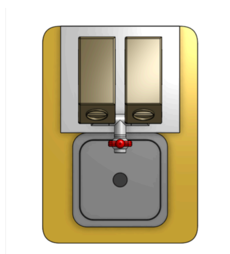
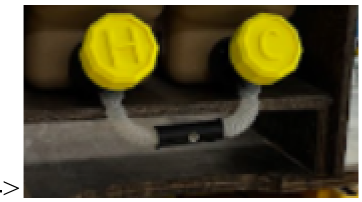
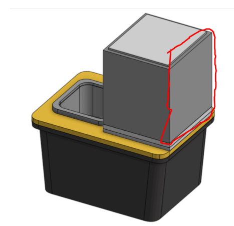
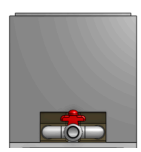
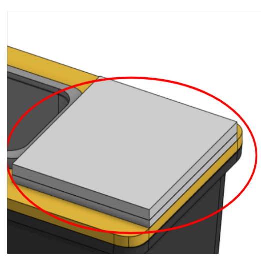
# Element G

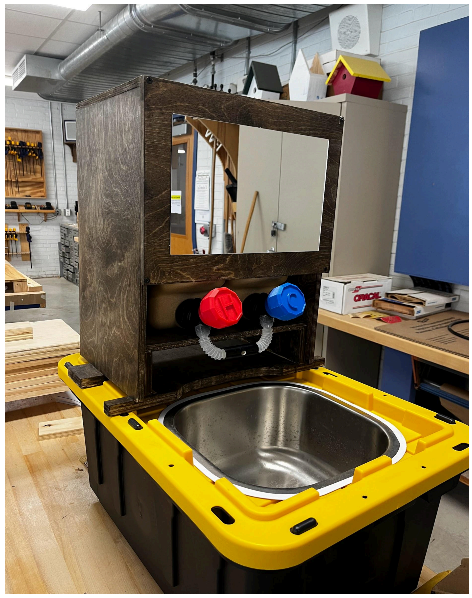
| Initial prototype (Brita Design)<br>January 1st- February 1st (2024) | Prototype going into Build<br>February 15th | Iterations Over Build Period February (images below include the changes)<br>15th-April 15th |
|---|---|---|
|  | <br><br> |  -> <br>(1)<br> -> <br>(2)<br> -> <br>(3)<br> -> <br>(4) |
| **Strategy:** Use Brita tanks to store | **Strategy**: Use the | **Strategy**: The iterations here were made to solve |

| | | |
|---|---|---|
| the water. Would allow conservation of water as the rinse-water could be reused . | ability to purchase PVC to attach water jugs to a valve that has a PVC adapter.; | problems during production or were found to better enhance the final iteration that is pictured to the right. |
| **Material needed specific to this iteration**<br><br>- Brita tanks<br>- Circular sink bowl<br>- 3-D printed handles<br>- Waste Container within the storage container | **Materials needed specific to this iteration:**<br>- Storage container<br>- Sink bowl<br>- Water jugs (2)<br>- L PVC connector (2)<br>- PVC Pipe (1ft)<br>- T PVC connector (1)<br>- PVC Valve (1)<br>- 15x16x.75 in. Wood Panels (4)<br>- 15x 13x.75 in. Wood Panels (4)<br>- | **Materials added or removed**<br><br>- 3-D Printed Sink Handles(added)<br>- PVC Connectors(removed)<br>- PVC Valve (removed)<br>- Hose tubing [included with purchase of water tanks] (added)<br>- Spray Paint [Black (added)<br>- Adhesive Mirror |
| **Pros**<br>- Had a promising capability to be portable based off compactability and convenient handles<br>- Used many materials that could be purchased at reasonable costs<br>**Topical Issues**<br><br>- The design was not thoroughly thought through in terms of how this prototype could be produced. For example, in thought, the handles attached to the container were convenient in carrying the sink, but there was not a solution as to how those handles could be secured to the container.<br>- Lack of true | **Pros**:<br>● Uses a standardized tubing system. PVC comes in standard sizes and therefore makes connections between PVC pieces easy<br>●<br>**Topical issues :**<br><br>- No definitive plan as to how to fasten the PVC tubing to the water containers without significantly altering the water tanks. | **Issues during Manufacturing**<br><br>● During the testing of the flow rate of the sink, the two plastic tubes attached to the short piece of PVC cracked and were deemed unusable. This caused the group to consider using a glue substance to secure these tubes to the two spigots to mitigate this recurrence.<br>**Specific Reasons for changes**<br>● (1) Adding Larger Sink Handles with a temperature distinction allows for a large surface to grip on to, as well as learning tools to understand different knobs provide different water temperatures. We also initially had the handles both the color yellow, however, they were remade as Blue and Red to distinguish temperatures for the children. [*Note: temperature of the water can only be controlled if the teacher adds different temperature water in each container. There is no internal water heating system or cooling system*] |

| | | |
|---|---|---|
| measurements to ensure each piece purchased could be integrated successfully. | - Unsure how to fasten the box containing the water tanks to the container during use | • (2) The original plan was to use the removability of the top piece of wood to have access to the water tanks. However, there was no functional way of completing this. The group decided the back was not essential to the functionality, and portability, and therefore chose to proceed with an open back. Further testing may occur by the Whitemarsh teachers that could cause a future iteration of this specific piece<br>• (3) Initially, the group did not agree on a mirror, and therefore went into the build not intending on implementing a mirror. However during the build process, it was felt that it ultimately would be a pleasing add on to have to better simulate a bathroom experience with a sink. Additionally, the original prototype included the front panel wrapping around the sink handles. This was changed due to an inability to efficiently cut this piece to specification on the CNC machine.<br>• (4) Originally the plan was to elevate the water containers above the sink bowl by placing planks of wood underneath the tanks. However, during manufacturing, the group determined this would add unnecessary weight, and therefore needed to create a new system of suspending the containers. Further, we came up with a box where there are spaces underneath. This not only used less wood, but also reduced the weight. |

Tools Used through process: CNC machine, Drills, Wood stain, paint brushes, palm sanders, table saw, clamps, cast saw, measuring tapes.

| |
|---|
| Final Prototype submitted to Whitemarsh Special Education Program* |



Final Material List

ORLANDO 13 x 13 inch Small Bar Sink Undermount Single Bowl Stainless Steel 18 Gauge Kitchen Sink With Basket Strainer - Amazon.com

HDX 27 Gal. Tough Storage Tote in Black with Yellow Lid HDX27GONLINE(5) - The Home Depot

.75 in Plywood: Supplied by school

3D- printed sink handles: Film provided by school

8 x 13 in  Adhesive mirror

.5 in PVC Pipe- Supplied by school

*No aspects of the final prototype were identified as unable to be evaluated either mathematically or physically prototyped. The group has carefully considered each piece, and will proceed with caution. If there are aspects that are found post-project, they will be revised and reported to the stakeholders.